CS 747: Programming Assignment 3

Margav Mukeshbhai Savsani

200050072

1. Initial Attempt

I implemented Linear Sarsa with tile encoding using a lot of features like speed, angle between target direction and heading direction, distances from target and the four walls (I thought this would help me generalise for task 2). What I realised is that the reward being returned was not that informative for the controller to learn too quickly and hence I used modified reward in my update conditions. But still I was not getting good performance and hence I got away with the rest of the features and just kept the angle between target direction and heading direction as a single feature. This performed better but was able to get only 1.45/2.5 in the autograder. The reason for this score is that the model is getting no episodes to learn and is being evaluated from the first episode itself. Had the weights being learned by making it run for a large number of episodes and then evaluated I believe it would have done its task. But since we were not to do it this way, I finally decided not to learn a model and put human intuition to take actions(I was already doing this by writing appropriate conditions to modify reward). My code for this initial attempt is here

2. Task 1: The Parking Lot Problem

I used angle between target direction and heading direction to decide my actions. The main logic is that I accelerate(acceleration action) my car only when this angle is within a bound(almost zero). My steer action is such that it tries to bring this angle to zero. Thus by using simple conditional statements I was able to do this task. This approach clearly imitates what any human would have done had he/she been driving that car.

3. Task 2: The Parking Lot Problem Intensifies

For task 2, I made use of the topology of the problem and realised that there are no obstacles near the y=0 line. So my car tries to reach y=0 line and after reaching there moves in a straight line to the target. The features that I made use of are x coordinates, y coordinates and the angle between target direction and heading direction. At the starting position I checked if it was possible(without hitting pits) to move vertically to reach y=0 line. If it was possible I simply made my car head vertically, move to y=0 and stop there, head horizontally towards target and then move towards the target. If the vertical movement was not possible I checked if it can move rightwards(which would always be true given the topology of the problem) and thus moved the car horizontally to a new position from where I repeated my procedure of checking if vertical movement is possible and so on. Thus by using conditional statements and some internal variables I was able to implement this task.